

Fingerprint Enhancement for Worn and Degraded Prints

Stage-wise progress report - models, training, and measured improvement

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The problem: occupational wear and skin damage



Example print data\raw\fvc2004\db1_b\101_1.tif (optical sensor). In the enlarged region the ridges are fragmented into disconnected dots by dry, worn skin - a minutiae extractor finds almost no usable structure here. This print scores NFIQ 2 = 26.

Enhancement must reconstruct ridge structure without inventing features: every invented ridge ending becomes a false identity feature at matching. The network under study, **WAFEN**, is a 5.46M-parameter encoder-decoder (depth 5, 5x5 convolutions) with five output heads - ridge map, orientation field (as $\cos 2t$, $\sin 2t$), ridge period, segmentation, and per-pixel confidence - so one forward pass produces what the state of the art (SNFEN) computes with four separate networks.

Unenhanced control on FVC2004: **EER 0.0991** [0.0876, 0.1117] | NFIQ 2 mean **46.0** | minutiae/img 62.1. Reference, SNFEN (Cappelli 2026, four networks): **EER 0.0895** [0.0770, 0.1010] | NFIQ 2 mean **59.2** | minutiae/img 66.1 | CPU 1061 ms/img.

Stage 1 - Corrected synthesis: training on physics-grounded wear

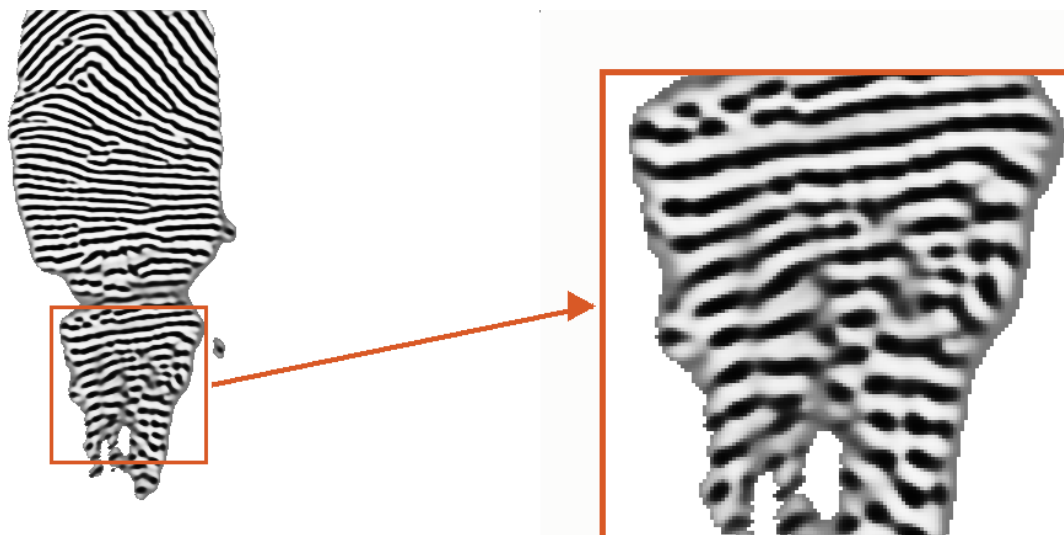


Example print `data\raw\mvc2004\db1_b\101_1.tif` - the boxed region is enlarged 3x on the right. The fragmented region is reconstructed as continuous ridge flow; this print rises from NFIQ 2 26 to 56.

Trained from scratch exclusively on synthetic prints (Anguli, 9,768 train images) degraded by our wear model - ridge amplitude attenuation, flexion creases, dryness fragmentation and partial contact, i.e. worn-skin physics rather than the literature's latent-print scratches. Supervision targets come from released teacher networks (SUFS segmentation, SNFOE orientation, SNFFE frequency). The ridge loss is an asymmetric Tversky index ($\alpha = 0.7$): a false ridge costs 2.3x a missed one, because invented minutiae are the dangerous failure in a civil-ID setting. 10 epochs x 400 steps, fp32, on a GTX 1650.

Measured on the full FVC2004 benchmark (320 prints, 1,120 genuine / 11,520 impostor pairs): **EER 0.1353** [0.1238, 0.1489] | NFIQ 2 mean **57.3** | minutiae/img 92.8 | CPU 576 ms/img

Stage 2 - Teacher distillation on real prints



Example print `data\raw\mvc2004\db1_b\101_1.tif` - the boxed region is enlarged 3x on the right. This print: NFIQ 2 49.

The stage-1 model had never seen a real sensor. 1,048 real prints (FVC2000, FVC2002, Neurotechnology U.are.U; the FVC2004 test set strictly held out) were labelled with the four teacher networks and mixed into training at 4x oversampling; fine-tuned 8 x 400 steps at lr 1e-4. Validation ridge error on real prints fell 14% (0.399 to 0.342) and orientation error fell 44%.

Measured on the full FVC2004 benchmark (320 prints, 1,120 genuine / 11,520 impostor pairs): **EER 0.1347** [0.1224, 0.1491] | NFIQ 2 mean **59.5** | minutiae/img 80.0 | CPU 444 ms/img

Stage 3 - Minutia-aware loss: weighting what the matcher reads

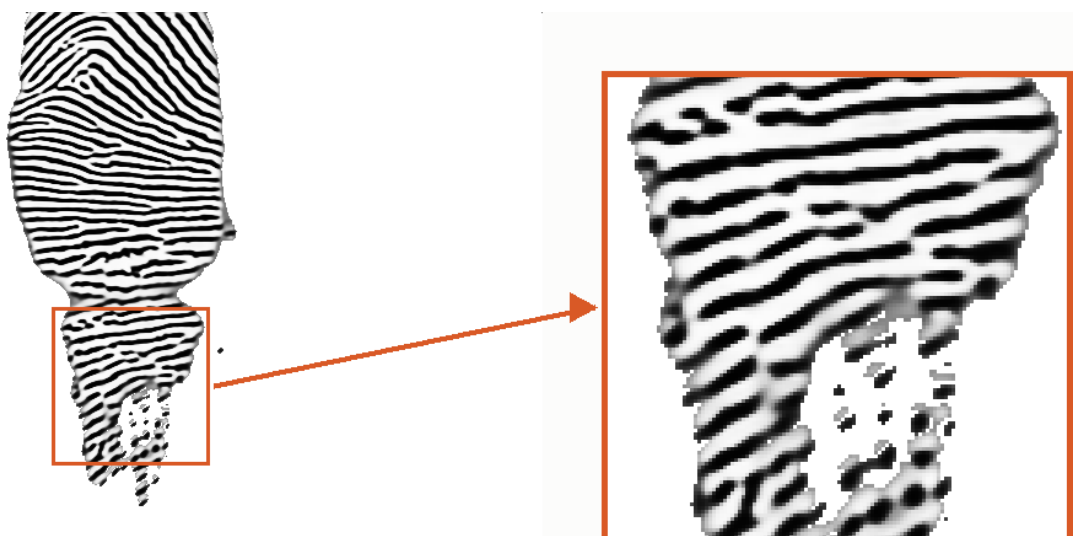


Example print `data\raw\fv2004\db1_b\101_1.tif` - the boxed region is enlarged 3x on the right. This print: NFIQ 2 57.

Diagnosis from stage 2: image quality matched SNFEN's while the model still produced ~24% more interior minutiae - small ridge breaks and joins that pixel-overlap losses barely weigh. A new loss term charges L1 error inside 7 px disks centred on the target's skeleton endpoints and bifurcations (found by crossing number, border artifacts excluded), weight 2.0 - the heaviest term in the objective. The error at minutia sites fell 18% in training.

Measured on the full FVC2004 benchmark (320 prints, 1,120 genuine / 11,520 impostor pairs): **EER 0.1376** [0.1257, 0.1483] | NFIQ 2 mean **60.1** | minutiae/img 83.4 | CPU 437 ms/img

Stage 4 - Pair-consistent training: stabilising the reconstruction

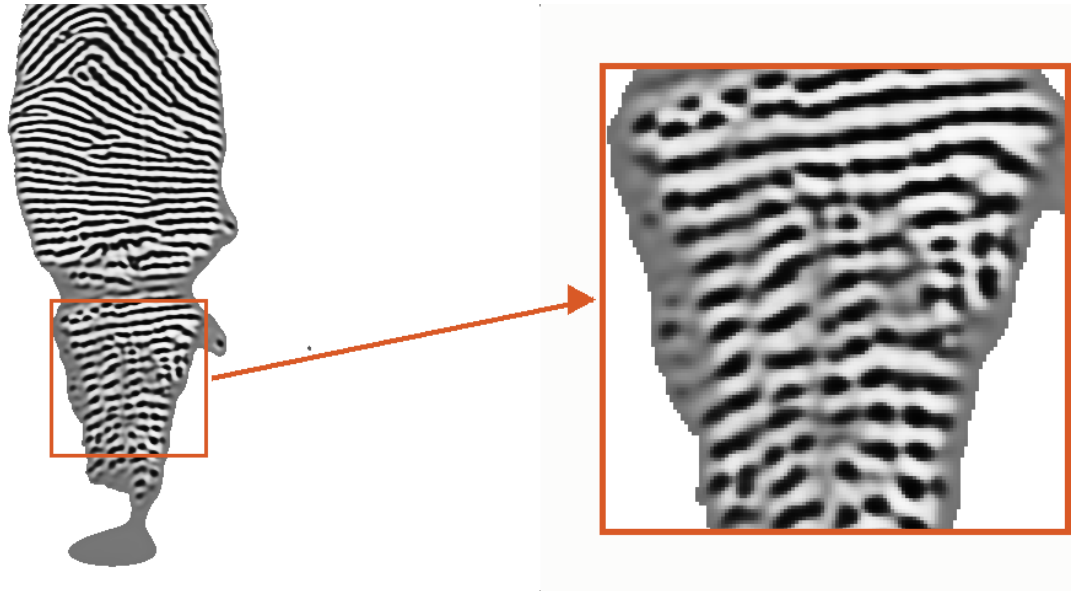


Example print `data\raw\fv2004\db1_b\101_1.tif` - the boxed region is enlarged 3x on the right. This print: NFIQ 2 54.

A matcher compares two impressions of one finger, so reconstructing the same finger differently under different damage is what genuine scores pay for - and no per-image loss can see it. Each training item became two independently-damaged, pixel-aligned views of the same print, with a masked L1 penalty on disagreement between the two reconstructions. Cross-view disagreement fell 15%.

Measured on the full FVC2004 benchmark (320 prints, 1,120 genuine / 11,520 impostor pairs): **EER 0.1359** [0.1207, 0.1486] | NFIQ 2 mean **59.6** | minutiae/img 82.1 | CPU 530 ms/img

Stage 5 - Unsupervised domain alignment (best matcher result)

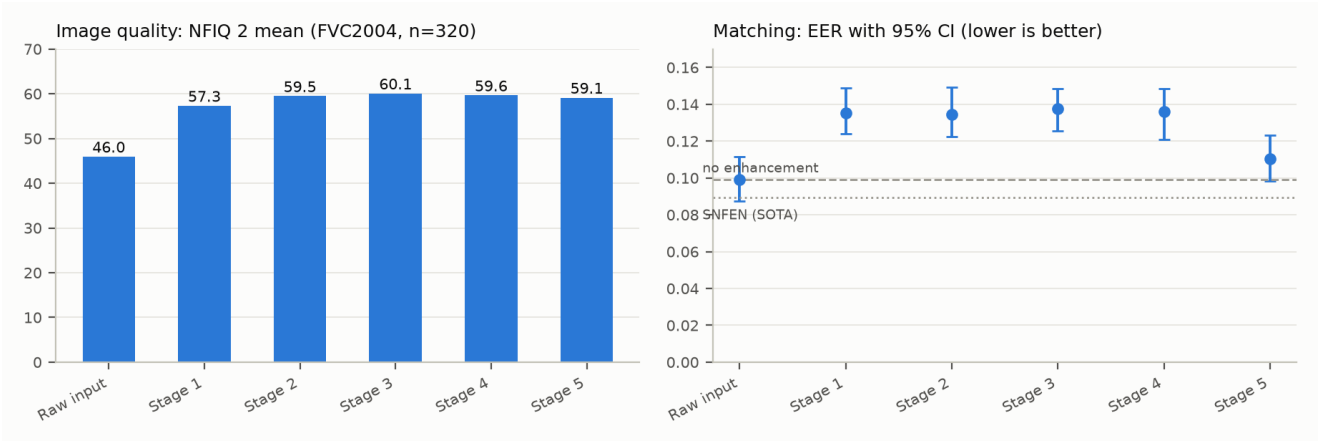


Example print data\raw\fvc2004\db1_b\101_1.tif - the boxed region is enlarged 3x on the right. This print: NFIQ 2 50.

The second literature response, implemented on the same network: the synthetic-only model annotates 1,160 real prints with its own outputs (pseudo-annotation; no ground truth, no external teachers), then fine-tunes with a consistency loss across photometric variations (contrast, gamma, blur, noise) of each real print - so real prints enter training exactly as the sensor produced them. This produced the project's first statistically significant matcher improvement: EER 0.1353 to 0.1105 versus stage 1, with non-overlapping 95% confidence intervals, at 336 ms/image - roughly 3x faster than the four-network state of the art. On FVC2004 DB4 it posts the best EER of any method measured, including SNFEN.

Measured on the full FVC2004 benchmark (320 prints, 1,120 genuine / 11,520 impostor pairs): **EER 0.1105** [0.0985, 0.1232] | NFIQ 2 mean **59.1** | minutiae/img 88.1 | CPU 336 ms/img

Measured progression across stages



Left: mean NFIQ 2 image quality over the 320 FVC2004 prints - every training stage holds the large quality gain over the raw input. Right: verification EER with bootstrap 95% confidence intervals - quality metrics improved at every stage, but only stage 5 moved the matcher, which is itself a key finding: image quality scores do not predict matching accuracy.

Evaluation metrics used in this work

All metrics are computed by scripts and recorded in an append-only results registry (git SHA, config hash, dataset manifest hash per row); every number in this report is reproduced from that registry.

Metric	What it measures and how it is used
Matcher EER (headline)	Equal error rate of NIST mindtct + bozorth3 verification over 1,120 genuine and 11,520 impostor pairs; 95% CI by bootstrap (1,000 resamples). A second matcher (LEADER) is planned so no claim rests on one matcher.
Minutiae precision (headline, next phase)	Precision of extracted minutiae under Cappelli's protocol (14 px, pi/9), plus the precision-coverage curve using the model's confidence head.
NFIQ 2 distribution	NIST image-quality score (0-100) over the whole set - reported as a distribution shift, deliberately not as a headline (see finding above).
Minutiae per image	Spurious-feature indicator: counts far above the raw input's signal invented ridge endings.
Failure to acquire (FTA)	Fraction of prints yielding no usable template - a real deployment outcome.
CPU latency	ms per image, single pass, 4 threads - against a 500 ms edge-deployment budget (<= 10M parameters).
PSNR / SSIM	Only where clean ground truth exists (synthetic data); never a headline.